

BST Vol 8 Ch 1 — Classical Mechanics Substrate Reading (v0.3, Wave 3)

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Vol 8 Chapter 1 — Classical Mechanics Substrate Reading

Headline result

Classical mechanics emerges from substrate D_{IV}^5 via: 1. $\hbar \rightarrow 0$ limit of QM (Vol 5 Lyra): classical phase space from substrate Hilbert space 2. Yang-Mills classical action (Vol 7 Ch 8): Lagrangian for fields 3. Conservation laws (T2473-T2475 Lyra Friday): energy + momentum + charge from Noether on substrate symmetry 4. Substrate dynamics (Vol 1 Ch 7): Hamilton + Schrödinger limits

Honest weak-foundation scope: Vol 8 paradoxically has weakest BST coverage among physics volumes. Standard classical mechanics is well-established; substrate-derivation of action principles + ergodic theorems + chaos theory requires multi-week theorem work.

Substrate framework anchors

- T2473 Energy Conservation via $SO_0(5,2)$ time-translation
- T2474 Momentum Conservation via coset translation
- T2475 Charge Conservation via $SO(2)$ isotropy
- Vol 7 Ch 8 Yang-Mills classical action precedent
- Vol 5 (QM Lyra) classical limit framework

Cross-volume dependencies

- Vol 1 Ch 7 (Dynamics)
- Vol 5 (QM, Lyra) — classical limit
- Vol 6 (Stat Mech, Lyra) — ergodic
- Vol 7 Ch 8 (Lagrangian EM) — action principle precedent

K-audit anchor

K229 Vol 8 Ch 1 K-audit pre-stage (Keeper pending).

Pedagogical walkthrough

Level 1

Classical mechanics is the physics of Newton's laws — pushing balls, pendulums, planets orbiting. BST predicts classical mechanics emerges as the “limit” of quantum mechanics when systems are big. The substrate's quantum framework becomes classical when $\hbar \rightarrow 0$.

Level 2

Classical mechanics emerges from substrate D_IV⁵ via: - $\hbar \rightarrow 0$ limit of QM - Conservation laws T2473-T2475 (Lyra Friday SP-31 #279) - Yang-Mills classical action

Vol 8 acknowledges weak-foundation: substrate-derivation of action principles + chaos theory requires multi-week theorem work beyond current state.

Level 3

Per $\hbar \rightarrow 0$ classical limit of substrate QM (Vol 5 Lyra): phase space (q, p) emerges from substrate Hilbert space coherent-state framework. Hamilton's equations $dq/dt = \partial H/\partial p$, $dp/dt = -\partial H/\partial q$ substrate-natural via Bergman propagator semiclassical limit.

Conservation laws inherit from substrate symmetries via T2473-T2475: - Energy: SO₀(5,2) time-translation invariance + Casimir centrality - Momentum: coset translation in $\mathfrak{m} \subset \mathfrak{so}(5,2)/(\mathfrak{so}(5) \oplus \mathfrak{so}(2))$ - Charge: SO(2) isotropy commutation

Honest scope: action principle + ergodic + chaos theory require Vol 6 (stat mech, Lyra) + Vol 1 Ch 7 dynamics multi-week theorem work for full D-tier.

Bibliography

1. H. Goldstein *Classical Mechanics* standard text.
2. T2473-T2475 (Lyra Friday): conservation laws.
3. Vol 5 (QM, Lyra) classical limit framework.